University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Digital Logic Design	**Course Code:** CMPE-221L
Assignment Type: Lab	**Dated:** 15th September 2026
Semester: 3rd	**Session:** 2025
Lab Number: 1	**CLOs to be covered: CLO 1, CLO 3**
Lab Title: Testing MSI Integrated Circuits	**Teacher Name:** Fareed Jafri

Lab Goals

- Add 4-bit binary numbers using an integrated circuit (CLO1, CLO3).
- Interface a seven segment display unit with a 4-bit binary input (CLO1, CLO3).

Equipment Required

- **ICs:** 7483 (1x), 7447 (1x), Seven segment display (1x)
- **Resistors:** 220 Ohms (7x)
- **Tools:** IC puller (1x), Nose pliers (1x), Wire stripper (1x)
- **Other:** Jumper wires (as needed)

Lab Evaluation

	CLOs	Maximum marks	Marks obtained
Task 1a	1	3	
Task 1b	3	2	
Task 2a	1	3	
Task 2b	3	2	
Total			**/10**

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric	CLOs
Completed	**100%**	L1	Completed the task correctly.	All
Great attempt	**75%**	L2	Almost completed the task, with minor deficiencies	2,3
Good attempt	**50%**	L3	Did not complete the task but managed to reasonably attempt it.	All
Basic attempt	**25%**	L4	Did not complete the task but managed to basically attempt it.	All
Did not attempt	**0%**	L5	Did not complete the task or attempt to do it.	All

Operate a 4-bit binary adder integrated circuit

Task 1a [CLO1, 3 Marks]

The 7483 IC is a 4-bit binary adder, which adds two 4-bit binary numbers to produce a 5-bit output. The pinout of the IC is shown in the 'Pin out' section at the end. Wire up the IC as follows. Switches S0 to S3 provide the first 4-bit binary input A. Switches S4 to S7 provide the second 4-bit binary output B. LEDs L0 to L4 record the 5-bit binary output C.

Task 1b [CLO3, 2 Marks]

Now fill in the following table using the circuit from Task 1a.

Binary input A	**Binary input B**	**Binary output C**
0011	0000	
0011	0001	
0011	0010	
0011	0011	
0011	0100	
0011	0101	
0011	0110	
0011	0111	
1000	0101	
1001	0101	
1010	0101	
1011	0101	
1100	0101	
1101	0101	
1110	0101	
1111	0101	

Interface a seven segment display with a 4-bit binary input circuit

Task 2a [CLO1, 3 Marks]

The 7447 IC is is a seven-segment display driver. The pinout of the 7447 IC and that of a seven-segment display unit are shown in the pinout section at the end. Wire up the IC as follows. Switches S0 to S3 provide a 4-bit binary input to the display driver. The driver interfaces to a seven segment display unit using 100 Ohm resistors.

Task 2b [CLO3, 2 Marks]

Now fill in the following table using the circuit from Task 2a.

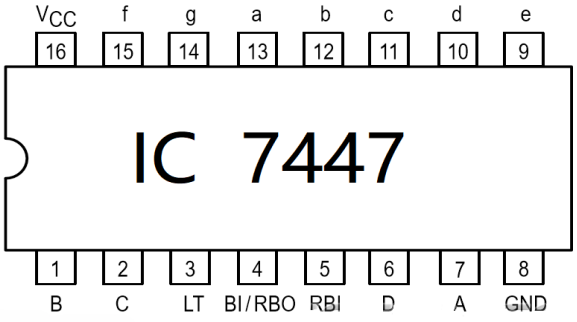
Binary input to seven-segment display driver	**Character displayed on seven-segment display unit**
0000	
0001	
0010	
0011	
0100	
0101	
0110	
0111	
1000	
1001	
1010	
1011	
1100	

Pinout

7483 IC



7447 IC



Seven-segment display

